THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Why does a person feel colder after bathing in air than in water? 1

(a) clothes are tight **(b)** water is cold
(c) evaporation takes heat away **(d)** air is wet

2. A student heats ice in a beaker with a thermometer. The temperature stays at 0 °C while the ice melts because: 1

(a) ice is cold **(b)** thermometer is wrong
(c) heater is off **(d)** heat is used to change the state

3. A student measures the length of a book as 24 cm. In metres it is: 1

(a) 24 m **(b)** 0.24 m **(c)** 2.4 m **(d)** 240 m

4. Which of the following would you choose for making a water bottle? 1

(a) paper **(b)** steel **(c)** cotton **(d)** cloth

5. In summer, a metal door handle feels hotter than a wooden door because: 1

(a) wood is colder
(b) wood is shiny
(c) the metal conducts heat to the hand faster
(d) metal is heavier

6. Why should we plant trees to prevent floods and droughts? 1

(a) trees give wood **(b)** trees are green
(c) trees are tall **(d)** roots hold soil and water

7. A thread 80 cm long is wound around a bottle exactly 4 times. The circumference of the bottle is: 1

(a) 4 cm **(b)** 320 cm **(c)** 80 cm **(d)** 20 cm

8. Which of the following would be a suitable material for making a window curtain which lets in some light? 1

(a) thin cotton cloth **(b)** brick
(c) thick metal sheet **(d)** wood

9. A student measures water temperature as 95 °C and later 60 °C. The heat flows from water to: 1

(a) water from surroundings **(b)** surroundings
(c) the thermometer to the water **(d)** nowhere

10. If the air is already full of water vapour, the evaporation rate: 1

(a) speeds up **(b)** stops completely always
(c) doubles **(d)** slows down

11. The Earth moving around the Sun is: 1

(a) linear **(b)** no motion
(c) orbital (revolution) **(d)** rotation only

12. A magnet can separate iron nails from: 1

(a) wooden chips **(b)** iron filings **(c)** nickel rods **(d)** steel pins

13. Why is it risky to put a mercury thermometer in the oven? 1

(a) it can break **(b)** it will show correct reading
(c) it becomes plastic **(d)** it is cold

14. The temperature of boiling water stays at 100 °C because: 1

(a) heat is used for changing water to steam **(b)** the beaker is hot
(c) thermometer breaks **(d)** water is cold

15. Length of a ribbon is 1.5 m. A boy needs pieces of 30 cm. How many pieces? 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Describe two uses of the thermos flask. 2

17. How does a water cooler or matka keep water cold? 2

18. A ribbon is 4.5 m long. Pieces of 25 cm are cut. How many can be made? 2

19. Why is steel preferred over iron for making bridges and cars? 2

20. Why should the bulb of a laboratory thermometer not touch the bottom of the beaker? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Design an experiment to compare the conduction of copper and iron rods. 3

22. Explain with an experiment that evaporation causes cooling. 3

23. Describe how a measuring tape is used to measure the waist of a person. 3

24. How can one prove that a cork floats because it is lighter than water and not because it is hollow? 3

OR

Explain why an iron nail sunk in hot water feels hot at once but a wooden stick feels warm slowly. 3

25. Explain how a sunny day and a cloudy day affect the drying of clothes. 3

OR

Which units would you choose for: length of a pen, length of a river, size of a bacterium? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A student measures the length of a table five times: 120.2, 120.4, 120.1, 120.3, 120.5 cm. Find the mean length and explain why repetition helps. 5

OR

Explain with experiments how the properties of materials (conduction, solubility, transparency, floating, hardness) can be tested. 5

27. Explain how a clinical thermometer works, why it has a kink and why it should not be heated. 5

OR

Explain the formation of rain from sea water step by step. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND